

Konv. radij

$$① R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

$$② \frac{1}{R} = \limsup_{n \rightarrow \infty} \sqrt[n]{a_n}$$

najv. stek.

6. Izračunaj R in obm. konv.

a) $\sum_{n=0}^{\infty} (n+1)x^n$ $R = \lim_{n \rightarrow \infty} \left| \frac{n+1}{n+2} \right| = 1 \Rightarrow$ konv. na $(-1, 1)$

Krajšica:

-1: $\sum_{n=0}^{\infty} (n+1)(-1)^n = +1 - 2 + 3 - 4 + \dots \Rightarrow$ ne konv., ker $\lim_{n \rightarrow \infty} (n+1) \neq 0$

1: $\sum_{n=0}^{\infty} (n+1) \Rightarrow$ div.

Konv. na $(-1, 1)$

b) $\sum_{n=1}^{\infty} \frac{x^n}{n^n}$ $\frac{1}{R} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{n^n}} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \rightarrow R = \infty$

Konv. na $(-\infty, \infty)$

c) $\sum_{n=1}^{\infty} \frac{x^{2n}}{4^n n} = \frac{1}{4}x^2 + \frac{1}{32}x^4 + \frac{1}{364}x^6 + \dots$

$$a_{2n} = \frac{1}{n \cdot 4^n}$$

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} \sqrt[n]{a_n} = \limsup_{n \rightarrow \infty} \sqrt[2n]{\frac{1}{4^n n}} = \frac{1}{2} \limsup_{n \rightarrow \infty} \sqrt[2n]{\frac{1}{4^n}} = \frac{1}{2}$$

$$a_{2n+1} = 0$$

$$R = 2$$

-2: $\sum_{n=1}^{\infty} \frac{(-2)^{2n} (4)^n}{4^n n} = \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow$ div

2: $\sum_{n=1}^{\infty} \frac{2^{2n}}{4^n n} = \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow$ div

konv. na $(-2, 2)$

7. Določí obrn. konv. in izračunaj vsoti!

a) $\sum_{h=1}^{\infty} \frac{(x-1)^h}{h}$ $R = \lim_{h \rightarrow \infty} \left| \frac{\frac{1}{h}}{\frac{1}{h+1}} \right| = 1 \rightarrow \text{konv. na } (0, 2)$

0: $\sum_{h=1}^{\infty} \frac{(-1)^h}{h}$ $\lim_{h \rightarrow \infty} \frac{1}{h} = 0 \Rightarrow \text{Leibniz} \Rightarrow \text{konv.}$

2: $\sum_{h=1}^{\infty} \frac{1^h}{h} \rightarrow \text{div}$

konv. na $[0, 2)$

$f'(x) = \sum_{h=1}^{\infty} \frac{1}{h} (x-1)^{h-1} =$

$= 1 + (x-1) + (x-1)^2 + \dots = \left(\frac{1}{1-(x-1)} \right) = \frac{1}{2-x}$

$f(x) = \int \frac{1}{2-x} dx = -\ln|2-x| + C$

$f(1) = 0 = -\ln 1 + C \rightarrow C = 0$

$f(x) = -\ln|2-x|$

b) $\sum_{h=1}^{\infty} \frac{n x^{h-1}}{(h+1)!} = f(x)$

$\lim_{h \rightarrow \infty} \left| \frac{\frac{h+1}{(h+2)!}}{\frac{h+2}{(h+3)!}} \right| = \lim_{h \rightarrow \infty} \left| \frac{(h+1)(h+3)}{(h+2)} \right| = \infty$

$a_n = \frac{(n+1)}{(n+2)!}$

$R = \infty$

$\int f(x) dx = \sum_{h=1}^{\infty} \frac{1}{(h+1)!} \left(\frac{x^h}{h} \right) + C = \sum_{h=1}^{\infty} \frac{x^h}{(h+1)!} + C = \frac{x}{2!} + \frac{x^2}{3!} + \dots =$

$e^x = 1 + x + \frac{x^2}{2} + \dots$

$= \frac{1}{x} \left(\frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) + C =$

$\underbrace{e^x - (1+x)}_{e^x - (1+x)}$

$= \frac{1}{x} (e^x - x - 1) + C$

$f(x) = -\frac{1}{x^2} (e^x - x - 1) + \frac{1}{x} (e^x - 1) = \frac{x e^x - x - e^x + 1}{x^2} = \underline{\underline{\frac{e^x(x-1) + 1}{x^2}}}$

8. Določi konv. obm. in izrač. vsoto

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \quad \frac{1}{R} \approx \lim_{n \rightarrow \infty} \sqrt[2n+1]{\frac{(-1)^n}{2n+1}} = 1$$

$$\underline{R=1}$$

$$-1: \sum \frac{(-1)^n}{2n+1} (-1) \rightarrow \text{konv}$$

1: podobno

konv. na $[-1, 1]$

$$f'(x) = \sum_{n=0}^{\infty} (-1)^n x^{2n} = \underbrace{1 - x^2 + x^4 - \dots}_{q = -x^2} = \frac{1}{x^2 + 1}$$

$$\frac{1}{1-q} = \frac{1}{1+x^2}$$

$$f(x) = \arctg x + C$$

$$f(0) = \arctg(0) + C = 0$$

$$C = 0$$

$$f(x) = \arctg x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \quad \text{na } x \in [-1, 1]$$

- In izračunaj vsoto

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \arctg 1 = \underline{\underline{\frac{\pi}{4}}}$$

$\nearrow$
 $x=1$

Weierstrassov krit. za en. konv. f_n

$$f: I \rightarrow \mathbb{R}$$

$$\epsilon_n = \sup_{x \in I} |f_n(x) - f(x)|$$

$$\bar{c} \in \lim_{n \rightarrow \infty} \epsilon_n = 0 \Rightarrow f_n \text{ konv. k } f$$

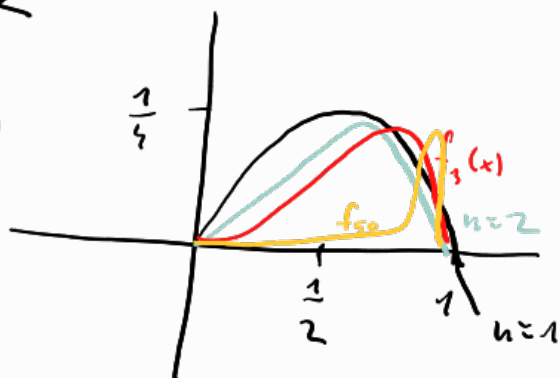
9. funkcijske vrste - določi limitno funk., ali je konv. enakomerno.

a) $f_n: [0, 1] \rightarrow \mathbb{R}$

$$f_n = n(1-x)x^n$$

$$f_1 = x - x^2$$

$$f_2 = 2x^2(1-x)$$



$$f'_n(x) = n(n x^{n-1} - (n+1)x^n) = n x^{n-1} (n - (n+1)x)$$

$$\max: n x^{n-1} = (n+1)x^n \quad x = \frac{n}{n+1}$$

$$f(x) = \lim_{n \rightarrow \infty} n(1-x)x^n = (1-x) \lim_{n \rightarrow \infty} \frac{n}{(\frac{1}{x})^n} = \lim_{n \rightarrow \infty} \frac{1}{(\frac{1}{x})^n \ln(\frac{1}{x})} = 0 \quad \forall x \in [0, 1]$$

$$f(x) = 0$$

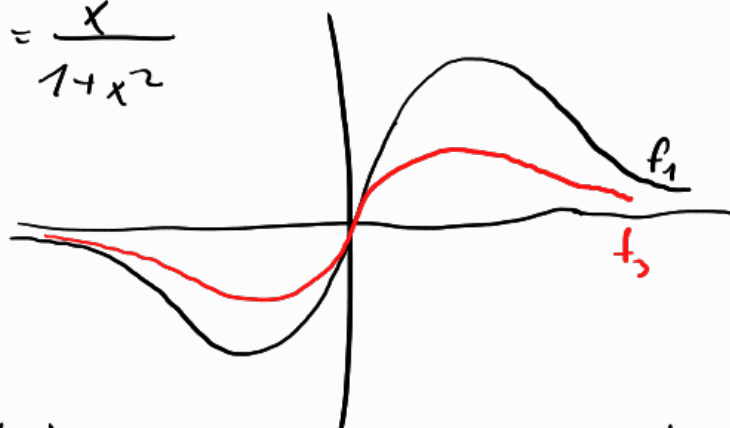
$$C_n = \sup_{x \in [0, 1]} |f_n(x) - f(x)| = \max_{x \in [0, 1]} |n(1-x)x^n| = \left| n \left(1 - \frac{n}{n+1}\right) \left(\frac{n}{n+1}\right)^n \right| = \left(\frac{n}{n+1}\right)^{n+1}$$

$$\lim_{n \rightarrow \infty} C_n = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^{n+1} = \lim_{n \rightarrow \infty} \left(1 + \frac{-1}{n+1}\right)^{n+1} = e^{-1} \Rightarrow \text{ne konv. enakom.}$$

b) $f_n(x) = \frac{x}{1+n^2 x^2}$

$$f_n: \mathbb{R} \rightarrow \mathbb{R}$$

$$f_1(x) = \frac{x}{1+x^2}$$



$$f(x) = \lim_{n \rightarrow \infty} \frac{x}{1+n^2 x^2} = 0 \quad \forall x \in \mathbb{R}$$

$$f'_1(x) = \frac{(1+n^2 x^2) - x \cdot 2n^2 x}{(1+n^2 x^2)^2}$$

$$f_n(\infty) = 0$$

$$f_n(0) = 0$$

$$\Rightarrow C_n = \sup_{x \in [0, \infty)} |f_n(x) - 0| = \frac{1}{n}$$

$$= \frac{1}{n} \rightarrow 0 \Rightarrow \text{weis. enakom. konv.}$$

$$\text{elstr.: } 1+n^2 x^2 - 2x^2 n^2 = 0$$

$$x^2 n^2 = 1 \quad x = \frac{1}{n} \quad \text{gledamo poz. ekstr.}$$

1.) Če f_n odvaja, $f_n \xrightarrow{\text{p. t.}} f$, $f'_n \in K$ na $I \Rightarrow f$ odv. 1

$$\lim f'_n(x) = f'(x)$$

2.) $f_n^{zv} \xrightarrow{K} f \Rightarrow \lim \int_a^b f_n(x) dx = \int_a^b f(x) dx$

Funkc. vrste

1.) $\sum_{n=1}^{\infty} f_n(x)$ konv. po točkah, če za $\forall x \in I: \sum_{n=1}^{\infty} f_n(x) < \infty$

2.) $\sum_{n=1}^{\infty} f_n(x) \in K$, če za $\forall \varepsilon > 0 \exists n_0 \in \mathbb{N}$, da za $\forall n > n_0$ velja $\left| \sum_{n=n_0}^{\infty} f_n(x) \right| < \varepsilon$ za $\forall x \in I$

Velja:

- Če f_n zvezna in $\sum f_n(x) \in K \Rightarrow F(x) = \sum f_n(x)$ zvezna

- Weierstrassov krit. za konv. funkc. vrst:

$$\sup_{x \in I} |f_n(x)| \leq c_n \quad \text{če} \quad \sum_{n=1}^{\infty} c_n < \infty \Rightarrow \sum_{n=1}^{\infty} f_n(x) \in K$$

10. Določi def. obm. in pokaži, da sta zvezni

a) $W(x) = \sum_{n=1}^{\infty} \frac{\sin(2^n x)}{2^n}$

$$W: \mathbb{R} \rightarrow \mathbb{R}$$

$$\sup \left| \frac{\sin(2^n x)}{2^n} \right| \leq \frac{1}{2^n} = c_n$$

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \rightarrow \text{geom. vrsta, } q < 1 \Rightarrow \text{konv.} \Rightarrow \sum_{n=1}^{\infty} \frac{\sin(2^n x)}{2^n} \in K \Rightarrow W(x) \text{ je zv.}$$

b) $S(x) = \sum_{n=1}^{\infty} \frac{1}{n^x}$

$$\sup_{x \in (1, \infty)} \left| \frac{1}{n^x} \right| \rightarrow \sup_{x \in [a, \infty)} \left| \frac{1}{n^x} \right| \leq \frac{1}{n^a} = c_n$$

$a > 1; [a, \infty) \subset (1, \infty)$

$$S: (1, \infty) \rightarrow \mathbb{R}$$

$$\sum c_n \text{ konv.} \Rightarrow \sum \frac{1}{n^x} \in K \Rightarrow S \text{ je zvezna na } [a, \infty)$$